

THE COMPLETE GUIDE

```
go func() {  
    ch <- work  
    }()  
    <-done
```

Mastering Go & Concurrency

From solid language foundations to true mastery of goroutines, channels, the scheduler, and the patterns that power production systems.

Goroutines

Channels

select

The G-M-P Scheduler

sync & atomic

context

Memory Model

Pipelines

Worker Pools

errgroup

Race Detector

Profiling

A structured path · 27 chapters · 4 parts · idiomatic, runnable Go

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Four parts, twenty-seven chapters — foundations through mastery

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How to Use This Guide

This is a structured path to **genuine mastery** of Go, with a deliberate, deep focus on concurrency — the feature that defines the language.

The guide is organized into four parts:

- **Part I — Go Foundations.** The language semantics you must hold in your head before concurrency makes sense: the type and memory model, interfaces, slices, and error handling. Tight but complete.
- **Part II — Concurrency Fundamentals.** Goroutines, the runtime scheduler, channels, `select`, the `sync` and `sync/atomic` packages, `context`, and the formal memory model. This is the core.
- **Part III — Concurrency Patterns.** The reusable structures real Go systems are built from: pipelines, fan-out/fan-in, worker pools, semaphores, rate limiting, cancellation, and `errgroup`.
- **Part IV — Mastery.** What separates someone who *uses* goroutines from someone who *reasons* about them: bug taxonomies, the race detector, profiling, tracing, tuning, and a dense idiom reference.

A note on philosophy. Go's concurrency model is built on a single famous idea, attributed to Rob Pike:

Do not communicate by sharing memory; instead, share memory by communicating.

You will see what that means concretely, when to follow it, and — importantly — when **not** to (sometimes a plain mutex is the correct, faster, clearer tool). Mastery is knowing the difference.

Every code sample is idiomatic, modern Go and is written to compile and run. Type them out. Run them under `-race`. Break them on purpose.

Part I — Go Foundations

1. The Go Philosophy

Go was designed at Google by Robert Griesemer, Rob Pike, and Ken Thompson to solve problems of *engineering at scale*: slow builds, uncontrolled dependencies, and the difficulty of writing correct concurrent software on multicore machines. Understanding the design values explains nearly every language decision.

Simplicity is a feature, not a limitation. Go deliberately omits inheritance, generics-everywhere, exceptions, and operator overloading (generics were eventually added in 1.18, but narrowly). The goal is that a large team can read each other's code with minimal surprise. When you feel a feature is "missing," the usual answer is that its absence removes a class of confusion.

Composition over inheritance. Go has no class hierarchies. Behavior is assembled by embedding types and satisfying interfaces. There is no `extends`.

Interfaces are satisfied implicitly. A type implements an interface simply by having the right methods — there is no `implements` keyword. This decouples the definition of behavior from the types that provide it, and it is the quiet engine behind much of Go's flexibility.

Concurrency is a first-class language concern, not a library bolted on. `go`, `chan`, and `select` are keywords. The runtime ships a scheduler that multiplexes potentially millions of lightweight goroutines onto a handful of OS threads.

The tooling is part of the language. `go build`, `go test`, `go fmt`, `go vet`, the race detector, and the profiler are not third-party afterthoughts. `gofmt` ended all formatting debates by fiat: there is one true format.

Errors are values. Go has no exceptions for ordinary control flow. Functions return errors, and you handle them explicitly. This is verbose and intentional — the cost of an error path is visible at every call site.

The rest of Part I assumes you can write basic Go. It focuses on the semantics that trip people up and that you *must* understand precisely before the concurrency chapters.

2. Types, Values & Memory

2.1 Value semantics by default

Go assignments and function arguments **copy** values. This is the single most important mental model for predicting behavior.

```
type Point struct{ X, Y int }

func main() {
    a := Point{1, 2}
    b := a           // b is a COPY of a
    b.X = 99
    fmt.Println(a.X, b.X) // 1 99 - a is untouched
}
```

When you pass a struct to a function, the function gets its own copy. To mutate the caller's value, pass a pointer:

```
func moveBad(p Point) { p.X += 10 } // mutates a copy, no effect
func moveGood(p *Point){ p.X += 10 } // mutates through the pointer

func main() {
    p := Point{}
    moveBad(p); fmt.Println(p.X) // 0
    moveGood(&p); fmt.Println(p.X) // 10
}
```

This matters enormously for concurrency: a copied value cannot be a shared-memory data race, but a pointer (or a slice, map, or channel — see below) shared across goroutines can be.

2.2 Reference-like types

A few built-in types contain internal pointers, so copying the *header* still shares the *underlying data*. These are **slices**, **maps**, **channels**, and **functions** (and pointers themselves). Copy a slice variable and both copies point at the same backing array:

```
s := []int{1, 2, 3}
t := s           // copies the slice HEADER (ptr,len,cap), shares backing array
t[0] = 99
fmt.Println(s[0]) // 99 - they alias the same array
```

This is why a slice or map shared between goroutines without synchronization is a classic data race, even though "you only copied a variable."

2.3 Zero values

Every type has a **zero value**, and Go guarantees variables are initialized to it — there is no uninitialized memory.

Type	Zero value
numeric (<code>int</code> , <code>float64</code> , ...)	<code>0</code>
<code>bool</code>	<code>false</code>
<code>string</code>	<code>""</code> (empty, not nil)
pointer, slice, map, channel, func, interface	<code>nil</code>
struct	each field set to its zero value

A well-designed Go type is **useful at its zero value**. `sync.Mutex`, `sync.WaitGroup`, `bytes.Buffer`, and `sync.Once` all work correctly with no initialization — `var mu sync.Mutex` is a ready-to-use lock. Aim for this in your own types.

2.4 nil is typed and subtle

`nil` is not a single value; it is the zero value of several distinct types, and it behaves differently for each:

- A **nil slice** has length 0; you can `append` to it, `range` it, and call `len`. It is the idiomatic empty slice.
- A **nil map** can be read (returns zero values) but **panics on write**. You must `make` a map before assigning.
- A **nil channel** blocks forever on send *and* receive (a property `select` exploits — see Chapter 10).
- Calling a method on a **nil pointer** is fine *if the method does not dereference it*.

```
var m map[string]int
fmt.Println(m["x"]) // 0 - reading nil map is OK
m["x"] = 1          // PANIC: assignment to entry in nil map
```

2.5 Constants and `iota`

Constants are arbitrary-precision until assigned to a typed variable. `iota` generates successive integers within a `const` block, ideal for enums:

```
type State int
const (
    StateIdle   State = iota // 0
    StateRunning           // 1
    StateStopped           // 2
)
```

2.6 Stack, heap, and escape analysis

You never call `malloc` / `free`. The compiler decides whether a value lives on the goroutine's **stack** (cheap, automatically reclaimed) or the **heap** (managed by the garbage collector) via *escape analysis*: if a value's lifetime provably ends with the function, it stays on the stack; if a reference to it "escapes" (is returned, stored in a global, captured by a goroutine that outlives the call, etc.), it moves to the heap.

```
go build -gcflags='-m' ./... # prints escape-analysis decisions
```

Returning a pointer to a local variable is perfectly safe in Go — the compiler simply heap-allocates it. Goroutine closures frequently force escapes; we will revisit this in the performance chapter.

3. Functions, Methods & Interfaces

3.1 Functions are first-class values

Functions can be stored in variables, passed as arguments, returned, and closed over surrounding state. Closures are the basis of many concurrency patterns — and the source of one of Go's most infamous bugs (the loop-variable capture, Chapter 23).

```
func adder() func(int) int {
    sum := 0
    return func(x int) int { sum += x; return sum } // closes over sum
}
func main() {
    a := adder()
    fmt.Println(a(1), a(2), a(3)) // 1 3 6 - state persists in the closure
}
```

3.2 Multiple return values and the error idiom

Go functions routinely return a result and an error. This is *the* error-handling mechanism.

```
func half(x int) (int, error) {
    if x%2 != 0 {
        return 0, fmt.Errorf("half: %d is odd", x)
    }
    return x / 2, nil
}
func main() {
    if v, err := half(7); err != nil {
        fmt.Println("error:", err)
    } else {
        fmt.Println("got:", v)
    }
}
```

The `comma, err` form is everywhere: map lookups (`v, ok := m[k]`), type assertions (`v, ok := i.(T)`), and channel receives (`v, ok := <-ch`) all return a second boolean/error you should check.

3.3 Methods and receivers: value vs pointer

A method is a function with a *receiver*. The receiver can be a value or a pointer, and the choice has real consequences.

```
type Counter struct{ n int }

func (c Counter) Get() int { return c.n } // value receiver: reads a copy
func (c *Counter) Inc()   { c.n++ }      // pointer receiver: mutates original
```

Rules of thumb that matter for correctness and concurrency:

- Use a **pointer receiver** if the method mutates the receiver, if the struct is large (avoid copying), or if *any* method needs a pointer receiver (keep the method set consistent).
- Use a **value receiver** for small, immutable-by-design types.
- **Never mix** value and pointer receivers on the same type without good reason.
- A type containing a `sync.Mutex` (or any `sync` type) **must** use pointer receivers — copying a locked mutex is a bug `go vet` will flag.

3.4 Interfaces: implicit, small, powerful

An interface is a set of method signatures. A concrete type satisfies it automatically by implementing those methods — no declaration required.

```
type Stringer interface {
    String() string
}

type Color struct{ R, G, B uint8 }
func (c Color) String() string {
    return fmt.Sprintf("#%02X%02X%02X", c.R, c.G, c.B)
}
// Color now satisfies fmt.Stringer with zero ceremony.
```

Keep interfaces small. The most powerful interfaces in the standard library have one method: `io.Reader`, `io.Writer`, `fmt.Stringer`, `error`. The Go proverb is: *"The bigger the interface, the weaker the abstraction."*

Accept interfaces, return structs. Functions should accept the narrowest interface they need (maximizing what callers can pass) and return concrete types (giving callers everything).

3.5 The interface value: (type, value) pair

This is essential for understanding `nil` checks and type assertions. An interface value is internally a pair: a **dynamic type** and a **dynamic value**. An interface is `nil` only when *both* are nil.

```
type myErr struct{}
func (*myErr) Error() string { return "boom" }

func doThing() error {
    var p *myErr = nil
    return p           // returns a non-nil interface holding (*myErr, nil)!
}

func main() {
    err := doThing()
    fmt.Println(err == nil) // false - the classic "typed nil" trap
}
```

The fix: return a literal `nil`, never a typed nil pointer, when you mean "no error." This bug surfaces constantly in error handling.

3.6 Type assertions and type switches

Extract the concrete type from an interface safely with the two-value form, or branch on type with a

`switch`:

```
func describe(i any) string {           // any == interface{}
    switch v := i.(type) {
    case int:    return fmt.Sprintf("int: %d", v)
    case string: return fmt.Sprintf("string: %q", v)
    case nil:   return "nil"
    default:    return fmt.Sprintf("unknown %T", v)
    }
}
```

4. Structs, Composition & Embedding

4.1 Composition, not inheritance

Go builds complex types by **embedding** other types. An embedded field's methods and fields are *promoted* to the outer type, giving inheritance-like ergonomics without the hierarchy.

```
type Engine struct{ Horsepower int }
func (e Engine) Start() string { return "vroom" }

type Car struct {
    Engine      // embedded (no field name)
    Brand string
}

func main() {
    c := Car{Engine{200}, "Tesla"}
    fmt.Println(c.Start())    // promoted from Engine
    fmt.Println(c.Horsepower) // promoted field
}
```

This is composition: `Car` has an `Engine` and borrows its behavior. There is no "is-a" relationship and no virtual dispatch — promotion is resolved at compile time.

4.2 Embedding interfaces

You can embed interfaces into interfaces to compose contracts — exactly how `io.ReadWriter` is built:

```
type Reader interface{ Read(p []byte) (int, error) }
type Writer interface{ Write(p []byte) (int, error) }
type ReadWriter interface {
    Reader
    Writer
}
```

4.3 Struct tags

Struct fields carry string *tags* read via reflection — the mechanism behind JSON, database, and validation libraries:

```
type User struct {
    ID    int    `json:"id"`
    Name  string `json:"name"`
    Email string `json:"email,omitempty"`
    Pwd   string `json:"- "` // never serialized
}
```

4.4 Comparability

Structs are comparable with `==` if *all* their fields are comparable. This lets structs be map keys. Slices, maps, and functions are **not** comparable (except to `nil`), so a struct containing one cannot be a map key or compared directly.

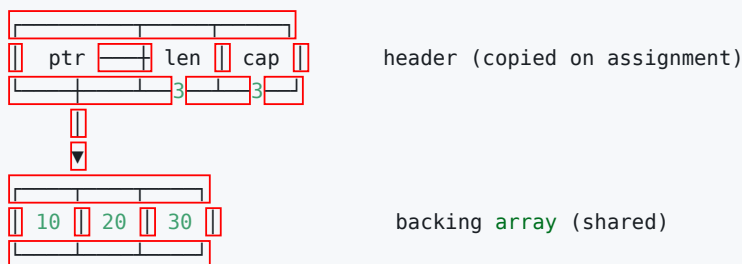
5. Slices, Maps & Their Internals

Understanding the runtime layout of slices and maps prevents a whole category of bugs — including subtle concurrency hazards.

5.1 The slice header

A slice is a small 3-word struct: a **pointer** to a backing array, a **length**, and a **capacity**.

```
slice := []int{10, 20, 30}
```



`len` is how many elements you can index; `cap` is how many fit before a reallocation is needed.

`append` writes into spare capacity if available; otherwise it allocates a *new, larger* backing array and copies.

5.2 The aliasing trap

Because slices share backing arrays, two slices can alias the same memory — re-slicing does **not** copy:

```
a := []int{1, 2, 3, 4, 5}
b := a[1:3]           // b shares a's array: len 2, cap 4
b[0] = 99
fmt.Println(a)        // [1 99 3 4 5] — a changed through b!
```

This is also a concurrency footgun: handing out sub-slices of a shared array to multiple goroutines shares memory between them.

5.3 The append reallocation surprise

Whether `append` mutates the original depends on spare capacity — making behavior *data-dependent* and easy to get wrong:

```
a := []int{1, 2, 3}    // len 3, cap 3 (typically)
b := append(a, 4)      // cap exceeded → NEW array; b is independent of a
```

```
x := make([]int, 3, 10) // len 3, cap 10
y := append(x, 99)      // spare capacity → SAME array; y[0..2] alias x
```

To force an independent copy, use `copy` or the three-index slice `a[low:high:max]` to cap capacity. To append-and-clone in one idiom:

```
dst := make([]int, len(src))
copy(dst, src)
// or: dst := append([]int(nil), src...)
```

5.4 Maps

Maps are hash tables holding key/value pairs. Critical properties:

- A `nil` map reads fine but **panics on write** — always `make(map[K]V)` first.
- **Iteration order is randomized** by design, to stop you depending on it. Sort keys if you need order.
- Maps are **not safe for concurrent use**: concurrent read+write (or write+write) is a data race that the runtime actively detects and crashes on with `fatal error: concurrent map ...`. For concurrent maps, use a `sync.Mutex` or `sync.Map` (Chapter 11).
- You cannot take the address of a map element (`&m[k]` is illegal), because rehashing may move it.

```
m := make(map[string]int)
m["a"]++           // works: missing key reads as 0, then stored as 1
v, ok := m["b"]    // ok == false, v == 0
delete(m, "a")     // remove
```

The `comma, ok` form is the only way to distinguish "key absent" from "key present with zero value."

6. Error Handling

6.1 Errors are values

`error` is just an interface:

```
type error interface {
    Error() string
}
```

Any type with an `Error() string` method is an error. Functions return errors; callers check them. There are no exceptions for normal control flow.

```
f, err := os.Open("config.yaml")
if err != nil {
    return fmt.Errorf("loading config: %w", err) // wrap and propagate
}
defer f.Close()
```

6.2 Wrapping with %w, and errors.Is / errors.As

`fmt.Errorf` with the `%w` verb **wraps** an error, building a chain that preserves the original while adding context. You then inspect the chain with `errors.Is` (match a sentinel) and `errors.As` (extract a concrete type).

```
var ErrNotFound = errors.New("not found")

func lookup(id int) error {
    return fmt.Errorf("lookup %d: %w", id, ErrNotFound)
}

func main() {
    err := lookup(42)
    fmt.Println(errors.Is(err, ErrNotFound)) // true - unwraps the chain

    var pe *os.PathError
    if errors.As(err, &pe) {                // true if any wrapped err is *os.PathError
        fmt.Println(pe.Path)
    }
}
```

Never compare errors with `==` unless you control the sentinel; use `errors.Is`. **Never string-match `err.Error()`** to branch logic — it is brittle.

6.3 Sentinel errors vs custom error types

- **Sentinel:** a package-level `var ErrX = errors.New(...)` for a well-known condition (`io.EOF`, `sql.ErrNoRows`). Match with `errors.Is`.
- **Custom type:** a struct implementing `error` when callers need structured data (a field, a code). Extract with `errors.As`.

```
type ValidationError struct {
    Field string
    Msg   string
}
func (e *ValidationError) Error() string {
    return fmt.Sprintf("%s: %s", e.Field, e.Msg)
}
```

6.4 The handle-or-propagate rule

At every error, do exactly one of: **handle it** (log, retry, default, recover) or **propagate it** (return, usually wrapped with context). Do not do both, and do not ignore it. The `errcheck` linter and `go vet` catch silent drops. Add context as you propagate so the final message reads like a stack trace: `loading config: open config.yaml: no such file or directory`.

7. defer, panic, and recover

7.1 defer

`defer` schedules a call to run when the surrounding function returns — used for guaranteed cleanup. Deferred calls run in **LIFO** order (last deferred, first executed).

```
func process(path string) error {
    f, err := os.Open(path)
    if err != nil {
        return err
    }
    defer f.Close()           // runs no matter how we return below

    mu.Lock()
    defer mu.Unlock()         // pairs lock with unlock; survives panics
    // ... work ...
    return nil
}
```

Argument evaluation is immediate; execution is deferred. Arguments to a deferred call are captured at the `defer` statement, not at execution:

```
func tricky() {
    i := 0
    defer fmt.Println("deferred:", i) // captures i == 0 NOW
    i = 99
    // prints "deferred: 0"
}
```

A deferred *closure*, by contrast, sees the final value because it reads the variable at run time:

```
defer func() { fmt.Println("deferred:", i) }() // prints 99
```

This distinction is the key to using `defer` with named return values to modify results or annotate errors.

7.2 panic and recover

`panic` aborts normal flow and unwinds the stack, running deferred functions along the way; if it reaches the top of a goroutine, the **program crashes**. `recover`, callable only inside a deferred function, stops the unwinding and returns the panic value.

```
func safeDivide(a, b int) (result int, err error) {
    defer func() {
        if r := recover(); r != nil {
            err = fmt.Errorf("recovered: %v", r)
        }
    }
}
```

```
    }()
    return a / b, nil // b == 0 panics; recover converts it to an error
}
```

Use panics sparingly. They are for *programmer errors* and truly unrecoverable states, not ordinary failures. Reserve them for: impossible conditions (`switch` default that "can't happen"), package initialization that cannot proceed, and converting to errors at a package boundary. Public APIs should return errors, not panic.

7.3 The cross-goroutine panic rule — critical for Part II

A panic does not cross goroutine boundaries. You cannot `recover` a panic that happens in a *different* goroutine. If any goroutine panics and is not recovered *within that same goroutine*, the **entire process crashes** — no matter how many `recover`s exist elsewhere.

```
func main() {
    defer func() { recover() }() // does NOT protect the goroutine below
    go func() {
        panic("boom") // crashes the whole program
    }()
    time.Sleep(time.Second)
}
```

Therefore: every goroutine that might panic must guard *itself*. This is why long-lived worker goroutines often wrap their body in a deferred `recover`. Keep this rule front of mind — it is one of the most common ways production Go services die.

Part II — Concurrency Fundamentals

8. Goroutines & the Runtime Scheduler

8.1 What a goroutine is

A **goroutine** is a function executing concurrently with other goroutines in the same address space. You start one by prefixing a call with `go`:

```
go doWork()           // runs doWork() concurrently; the go statement returns immediately
go func() {           // anonymous goroutine
    fmt.Println("hi from a goroutine")
}()
```

The `go` statement **does not wait**. It launches the goroutine and execution continues on the next line. Coordinating *when* goroutines finish is your job (Chapters 9–14).

8.2 Goroutines are not threads

This distinction is the heart of Go's concurrency story.

	OS thread	Goroutine
Initial stack	~1–8 MB, fixed	~2 KB, grows/shrinks dynamically
Created/destroyed by	kernel (expensive syscall)	Go runtime (cheap, user space)
Scheduled by	OS kernel	Go runtime scheduler
Context switch cost	~1–2 μ s (kernel involvement)	~tens of ns (no kernel)
Practical count	thousands	millions

Because a goroutine starts with a tiny stack that the runtime resizes on demand, you can run hundreds of thousands of them. Spawning a goroutine per incoming request or per connection is a normal, encouraged pattern — something that would be reckless with OS threads.

```
// Launch 100,000 goroutines – entirely reasonable in Go.
var wg sync.WaitGroup
for i := 0; i < 100_000; i++ {
    wg.Add(1)
    go func() {
        defer wg.Done()
        _ = i * i
    }()
}
wg.Wait()
```


8.3 The G-M-P scheduler model

The Go runtime contains a **work-stealing scheduler** that multiplexes goroutines onto OS threads. It is built from three entities — **G, M, and P**:

- **G (Goroutine)** — a goroutine: its stack, instruction pointer, and scheduling state.
- **M (Machine)** — an OS thread. Ms actually execute code. The runtime creates and parks them as needed.
- **P (Processor)** — a *logical* processor: a scheduling context holding a **local run queue** of runnable Gs. A P is the **permission to run Go code**. An M must hold a P to execute goroutines.

The number of Ps is set by `GOMAXPROCS` (default: the number of CPU cores). This is the key knob: it caps how many goroutines run *truly in parallel*.

```
GOMAXPROCS = number of P's = max parallelism
```

▼

each P is attached to an M (OS thread) to actually run a G

Idle P with empty queue? → STEAL half the G's from another P's queue.

How it fits together:

1. Each P keeps a **local queue** of runnable goroutines (fast, lock-free in the common case).
2. To run goroutines, the scheduler pairs a P with an M. The M runs goroutines from that P's queue.
3. When a P's local queue empties, its M tries to **steal** half the goroutines from another P's queue — *work stealing* keeps all cores busy and balances load automatically.
4. A **global queue** holds overflow and newly created goroutines that don't fit locally; Ps periodically check it (and are forced to, occasionally, for fairness).

8.4 Cooperative + preemptive scheduling

Goroutines yield the M at well-defined **scheduling points**: channel operations, `select`, blocking syscalls, `time.Sleep`, mutex contention, memory allocation, and function-call preludes. Historically a tight CPU loop with *no* such points could starve the scheduler. Since Go 1.14 the runtime uses **asynchronous preemption** (signal-based) so even a `for {}` loop can be preempted, preventing one goroutine from monopolizing a P.

8.5 Blocking syscalls and M handoff

When a goroutine makes a **blocking system call** (e.g., reading a file), its M blocks in the kernel. The runtime **detaches the P** from that blocked M and hands the P to another M (creating one if needed) so the other goroutines on that P keep running. When the syscall returns, the goroutine is rescheduled. This is why blocking I/O in one goroutine does not stall unrelated goroutines.

Network I/O is even better: it goes through the runtime's **netpoller** (epoll/kqueue/IOCP), so a goroutine blocked on a socket is *parked* without consuming an M at all. You write simple blocking-style code; the runtime makes it event-driven underneath.

8.6 GOMAXPROCS in practice

```
runtime.GOMAXPROCS(0)           // returns current setting (0 = query only)
n := runtime.NumCPU()           // logical cores available
runtime.NumGoroutine()          // live goroutine count – great for leak detection
```

Set it via the `GOMAXPROCS` environment variable or the function. **Container caveat (very common in production):** the Go runtime traditionally read the *host's* CPU count, not the container's CFS quota. A pod limited to 2 cores on a 64-core node could default `GOMAXPROCS` to 64, causing thrashing. The standard fix has been Uber's `automaxprocs` library, which sets `GOMAXPROCS` from the cgroup quota; recent Go versions have moved toward becoming cgroup-aware by default. When you deploy in containers, verify what `GOMAXPROCS` actually resolves to.

8.7 The mental model to carry forward

- A goroutine is *cheap*; do not pool them the way you would threads. Spawn freely, but always know how each one **stops** (Chapter 20).
- `GOMAXPROCS` Ps means *up to* that many goroutines run in parallel; the rest are *concurrent* — interleaved, not simultaneous.
- **Concurrency is not parallelism.** Concurrency is *structuring* a program as independently executing pieces; parallelism is *executing* pieces simultaneously on multiple cores. Go gives you concurrency as a design tool; the runtime delivers parallelism when hardware allows. A well-structured concurrent program runs correctly on one core and faster on many — without code changes.

9. Channels: The Complete Picture

Channels are typed conduits that let goroutines **communicate and synchronize**. They are the embodiment of "*share memory by communicating*." A channel both transfers a value and establishes a happens-before ordering between sender and receiver (Chapter 14).

9.1 Creating and using channels

```
ch := make(chan int)           // unbuffered channel of int
ch <- 42                       // SEND: blocks until a receiver is ready
v := <-ch                     // RECEIVE: blocks until a value is available
v, ok := <-ch                 // ok == false if channel is closed AND drained
close(ch)                     // close: no more sends allowed
```

9.2 Unbuffered channels: synchronous rendezvous

An **unbuffered** channel has no capacity. A send blocks until *some* goroutine receives; a receive blocks until *some* goroutine sends. The send and receive happen at the same instant — a **rendezvous**. This makes unbuffered channels a synchronization primitive as much as a data pipe: a successful send proves a receiver ran.

```
done := make(chan struct{})
go func() {
    fmt.Println("working...")
    close(done)           // signal completion
}()
<-done                   // blocks until the goroutine closes done
fmt.Println("worker finished")
```

`chan struct{}` is the idiomatic "signal only, no data" channel — `struct{}` occupies zero bytes.

9.3 Buffered channels: capacity and decoupling

A **buffered** channel holds up to N values. Sends block only when the buffer is *full*; receives block only when it is *empty*. This decouples sender and receiver timing.

```
ch := make(chan int, 3)      // capacity 3
ch <- 1; ch <- 2; ch <- 3     // all succeed without a receiver
// ch <- 4                  // would block: buffer full
fmt.Println(len(ch), cap(ch)) // 3 3 (len = queued, cap = capacity)
```

When to buffer. Use a buffer when you have a known, bounded amount of work to enqueue, or to smooth bursty producers, or to permit a fixed number of in-flight operations (a semaphore, Chapter 18). **Do not** reach for a buffer to "fix" a deadlock or a race — that usually hides a design flaw. Default to unbuffered; add a buffer only when you can explain *why* that size.

9.4 Closing channels — rules that prevent panics

`close(ch)` marks a channel as finished. The rules are strict, and violating them panics:

- **Sending on a closed channel panics.** ("send on closed channel")
- **Closing a closed channel panics.**
- **Closing a nil channel panics.**
- **Receiving from a closed channel never blocks:** it drains remaining buffered values, then returns the zero value forever, with `ok == false`.

The governing principle: **only the sender closes, and only when there is a single sender (or a coordinated owner)**. A receiver must never close a channel — it cannot know if the sender will send again. With multiple senders, do *not* have them close; use a separate signal (a `done` channel or `context`) to tell them to stop, and close downstream only after all senders have exited (often coordinated via a `WaitGroup`).

```
// Idiomatic: producer owns and closes; consumer ranges to completion.
func producer(ch chan<- int) {
    defer close(ch)           // sender closes when done
    for i := 0; i < 5; i++ {
        ch <- i
    }
}

func main() {
    ch := make(chan int)
    go producer(ch)
    for v := range ch {       // ranges until ch is closed AND drained
        fmt.Println(v)
    }
}
```

9.5 Ranging over a channel

`for v := range ch` receives values until the channel is **closed and drained**, then ends the loop. If the channel is never closed, the loop blocks forever — a frequent cause of goroutine leaks and `fatal error: all goroutines are asleep - deadlock!`.

9.6 Directional channel types

Restrict a channel's direction in function signatures to encode intent and let the compiler enforce it:

```
func produce(out chan<- int) { out <- 1 } // send-only: can only send
func consume(in <-chan int) { <-in }     // receive-only: can only receive
```

A bidirectional `chan int` converts implicitly to either directional form when passed. This is excellent self-documentation: a `chan<- T` parameter announces "I will only write," a `<-chan T` announces "I will only read." Pipelines (Chapter 15) lean heavily on this.

9.7 The nil-channel superpower

Operations on a **nil** channel block *forever*. This sounds useless but is the key to dynamically disabling a `select` case (Chapter 10): set a channel variable to `nil` and its branch becomes unreachable until you reassign it.

Channel state	Send	Receive
nil	block forever	block forever
open, empty	block (unbuffered) / proceed if buffer has room	block until a value arrives
open, has value/room	proceed	proceed
closed	panic	return zero, <code>ok=false</code> , immediately

Internalize this table — almost every channel bug is explained by one of its cells.

9.8 A complete worked example

```
func main() {
    jobs := make(chan int, 5)
    results := make(chan int, 5)

    // worker receives jobs, sends squares
    go func() {
        for j := range jobs {           // ends when jobs is closed
            results <- j * j
        }
        close(results)                 // safe: this goroutine is the only sender on results
    }()

    for i := 1; i <= 5; i++ {
        jobs <- i
    }
    close(jobs)                        // tells the worker no more jobs

    for r := range results {           // ends when results is closed
        fmt.Println(r)
    }
}
```

Notice the ownership discipline: each channel has exactly one closer, and closing flows *downstream* (close `jobs` → worker drains → worker closes `results` → main drains). This pattern scales to arbitrarily complex pipelines.

10. The select Statement

`select` is to channels what `switch` is to values: it waits on **multiple channel operations** and proceeds with whichever is ready. It is the control structure that makes channels composable.

10.1 Basic semantics

```
select {
  case v := <-ch1:
    fmt.Println("from ch1:", v)
  case ch2 <- 42:
    fmt.Println("sent to ch2")
  case <-done:
    fmt.Println("done signalled")
}
```

Rules:

- `select` blocks until **at least one** case can proceed.
- If **multiple** cases are ready, one is chosen **uniformly at random** — you cannot rely on ordering (this randomness deliberately prevents starvation).
- A `default` case makes the `select` **non-blocking**: if no other case is ready, `default` runs immediately.

10.2 Non-blocking operations with default

```
// Non-blocking receive
select {
  case v := <-ch:
    fmt.Println("got", v)
  default:
    fmt.Println("nothing ready")    // runs instead of blocking
}

// Non-blocking send (drop if no receiver / buffer full)
select {
  case ch <- v:
    // sent
  default:
    // dropped – useful for "best effort" metrics, lossy buffers
}
```

10.3 Timeouts with time.After

Combine `select` with `time.After` to bound how long you wait:

```
select {
case res := <-resultCh:
    fmt.Println("result:", res)
case <-time.After(2 * time.Second):
    fmt.Println("timed out")
}
```

Caveat: `time.After` allocates a new timer each call; in a hot loop, prefer a reusable `time.NewTimer` / `time.Ticker` and `Stop` it, or use `context.WithTimeout` (Chapter 13), to avoid timer churn. In modern Go the underlying timer is garbage-collected even if not drained, but explicit timers remain clearer for repeated use.

10.4 The done-channel cancellation pattern

The foundational cancellation idiom, and the conceptual ancestor of `context`. A goroutine selects on its work *and* a `done` channel; closing `done` unblocks every receiver simultaneously:

```
func worker(done <-chan struct{}, jobs <-chan int) {
    for {
        select {
        case <-done:                // cancellation wins
            fmt.Println("worker: shutting down")
            return
        case j, ok := <-jobs:
            if !ok {
                return            // jobs closed
            }
            fmt.Println("processing", j)
        }
    }
}

func main() {
    done := make(chan struct{})
    jobs := make(chan int)
    go worker(done, jobs)
    jobs <- 1
    jobs <- 2
    close(done)                // every worker's <-done fires at once
    time.Sleep(100 * time.Millisecond)
}
```

Closing a channel is a **broadcast**: any number of goroutines blocked on `<-done` all wake. This is why `done` channels are closed, not sent to — one `close` cancels N workers.

10.5 Disabling cases with nil channels

Because a nil channel blocks forever, assigning `nil` to a channel variable **removes** its case from contention. This lets you write a single `select` loop whose active cases change over time — for example, stop receiving once a source is exhausted while still draining an output:

```

func merge(in1, in2 <-chan int, out chan<- int) {
    for in1 != nil || in2 != nil {           // loop until both are disabled
        select {
            case v, ok := <-in1:
                if !ok {
                    in1 = nil                // disable this case; don't spin on a closed chan
                    continue
                }
                out <- v
            case v, ok := <-in2:
                if !ok {
                    in2 = nil
                    continue
                }
                out <- v
        }
    }
    close(out)
}

```

Without the `in1 = nil` trick, a closed channel is *always ready* and returns immediately, so the `select` would busy-loop at 100% CPU. Disabling the case is the correct fix. This is one of the most elegant and least-known tools in Go concurrency.

10.6 Empty select and select{}

`select {}` with no cases blocks **forever** — occasionally used to park `main` while background goroutines (e.g., an HTTP server's handlers) do the real work. Use sparingly; usually a `WaitGroup` or signal channel expresses intent better.

11. The sync Package

Channels are not always the right tool. When goroutines genuinely share state — a counter, a cache, a config — a lock is often simpler, faster, and clearer than routing everything through a channel. The Go proverb is honest about this: "*Channels orchestrate; mutexes serialize.*" Use channels to pass ownership and coordinate; use `sync` primitives to guard shared memory.

11.1 sync.Mutex — mutual exclusion

A `Mutex` ensures only one goroutine holds the lock at a time, serializing access to a *critical section*.

```
type SafeCounter struct {
    mu sync.Mutex
    n  int
}

func (c *SafeCounter) Inc() {
    c.mu.Lock()
    defer c.mu.Unlock()    // unlock even if the body panics
    c.n++
}

func (c *SafeCounter) Value() int {
    c.mu.Lock()
    defer c.mu.Unlock()
    return c.n
}
```

Essential rules:

- The zero value of a `Mutex` is an **unlocked, ready** mutex — never copy one after first use. A struct embedding a `Mutex` must be passed by pointer. `go vet` flags accidental copies.
- **Lock / Unlock must pair.** `defer mu.Unlock()` immediately after `Lock()` is the safest idiom; it survives panics and early returns.
- A Go `Mutex` is **not reentrant** — locking it twice from the same goroutine deadlocks. There is no recursive mutex; if you want one, your design is probably wrong. Refactor so the lock is taken once.
- Keep critical sections **short**. Never block on I/O, channels, or `time.Sleep` while holding a lock — you serialize everything behind it.

11.2 sync.RWMutex — many readers or one writer

When reads vastly outnumber writes, an `RWMutex` lets *many* readers proceed concurrently while writers get exclusive access.

```
type Cache struct {
    mu sync.RWMutex
    m  map[string]string
}

func (c *Cache) Get(k string) (string, bool) {
```

```

    c.mu.RLock()           // shared read lock – many can hold simultaneously
    defer c.mu.RUnlock()
    v, ok := c.m[k]
    return v, ok
}
func (c *Cache) Set(k, v string) {
    c.mu.Lock()           // exclusive write lock – blocks all readers and writers
    defer c.mu.Unlock()
    c.m[k] = v
}

```

`RWMutex` is **not always faster** than a plain `Mutex`: it has higher overhead per operation and can starve writers under heavy read load. Use it only when you have measured a read-heavy workload with non-trivial critical sections. For tiny critical sections, a plain `Mutex` often wins.

11.3 sync.WaitGroup — wait for a set of goroutines

A `WaitGroup` waits for a collection of goroutines to finish. `Add(n)` increases the counter, `Done()` decrements it (usually deferred), and `Wait()` blocks until it reaches zero.

```

func main() {
    var wg sync.WaitGroup
    for i := 0; i < 5; i++ {
        wg.Add(1)           // increment BEFORE launching
        go func() {
            defer wg.Done()  // decrement when the goroutine exits
            fmt.Println("working", i)
        }()
    }
    wg.Wait()              // block until all five call Done()
    fmt.Println("all done")
}

```

Rules that prevent the classic bugs:

- Call `Add` before the `go` statement, in the parent goroutine. Calling `Add` inside the new goroutine races with `Wait` and can let `Wait` return early.
- The counter must **net to zero**. Too many `Done`s panic (`negative WaitGroup counter`); too few make `Wait` block forever.
- **Do not reuse** a `WaitGroup` for a new batch until `Wait` has fully returned.
- Since Go 1.21, `sync.OnceFunc` / `OnceValue` and the `WaitGroup` API are stable; in modern Go you can also reach for `errgroup` (Chapter 21) when goroutines can fail.

11.4 sync.Once — exactly-once initialization

`Once` guarantees a function runs **exactly once**, even under concurrent calls — the idiomatic lazy singleton/initializer.

```

var (
    once sync.Once

```

```

    instance *Config
)
func GetConfig() *Config {
    once.Do(func() {
        instance = loadConfig()    // runs once; other callers block until it returns
    })
    return instance
}

```

All callers of `once.Do` block until the first invocation completes, so they all observe the fully initialized result (a happens-before guarantee). Go 1.21 added ergonomic helpers: `sync.OnceFunc(f)`, `sync.OnceValue(f)`, and `sync.OnceValues(f)` return memoizing functions and read better than a manual `Once` + package var.

11.5 sync.Cond — condition variables

`Cond` coordinates goroutines waiting for a *condition* to become true. It is rarely needed (channels usually express the same intent more clearly) but is the right tool when many goroutines wait on shared state guarded by a lock.

```

type Queue struct {
    mu    sync.Mutex
    cond  *sync.Cond
    items []int
}

func NewQueue() *Queue {
    q := &Queue{}
    q.cond = sync.NewCond(&q.mu)
    return q
}

func (q *Queue) Push(x int) {
    q.mu.Lock()
    q.items = append(q.items, x)
    q.mu.Unlock()
    q.cond.Signal()    // wake one waiter
}

func (q *Queue) Pop() int {
    q.mu.Lock()
    defer q.mu.Unlock()
    for len(q.items) == 0 {    // ALWAYS wait in a loop (guard against spurious wakeups)
        q.cond.Wait()        // atomically unlocks, sleeps, re-locks on wake
    }
    x := q.items[0]
    q.items = q.items[1:]
    return x
}

```

Key facts: `Wait` must be called with the lock **held**; it releases the lock while sleeping and reacquires it before returning. Always re-check the condition in a `for` loop, never an `if`. `Signal` wakes one waiter; `Broadcast` wakes all.

11.6 sync.Pool — reusing allocations

`Pool` is a free-list of temporary objects, used to reduce GC pressure in hot paths by recycling allocations (e.g., buffers).

```
var bufPool = sync.Pool{
    New: func() any { return new(bytes.Buffer) },
}
func process(data []byte) {
    buf := bufPool.Get().(*bytes.Buffer)
    buf.Reset() // ALWAYS reset — pooled objects are dirty
    defer bufPool.Put(buf)
    buf.Write(data)
    // ... use buf ...
}
```

Critical caveats: objects in a `Pool` may be **removed at any time** (notably during GC) — never assume an item persists. Always **reset** state before use; a pooled object retains whatever the last user left. `Pool` shines only when allocation/GC is a proven bottleneck; otherwise it adds complexity for no gain.

11.7 sync.Map — a concurrent map for specific shapes

The built-in `map` is unsafe for concurrent use. `sync.Map` is safe without external locking, but it is **specialized**, not a general replacement. It is optimized for two scenarios: (1) keys written once and read many times (a stable cache), or (2) disjoint goroutines touching disjoint key sets. For general read/write concurrency, a plain `map` guarded by a `Mutex` / `RMutex` is usually **faster and clearer**.

```
var m sync.Map
m.Store("a", 1) // write
v, ok := m.Load("a") // read
m.LoadOrStore("b", 2) // get-or-set atomically
m.Delete("a")
m.Range(func(k, v any) bool { // iterate; return false to stop
    fmt.Println(k, v)
    return true
})
```

Note the loss of type safety (`any` keys and values) and the absence of a `len`. Default to a guarded plain map; reach for `sync.Map` only when profiling shows lock contention and your access pattern matches one of the two it is built for.

11.8 Choosing: channel or mutex?

A practical decision guide:

- Use a **mutex** when goroutines share a piece of state and you are protecting *access to memory*: counters, caches, in-memory maps, config. Simple, fast, local.

- Use a **channel** when you are *transferring ownership* of data, distributing units of work, or *signaling events* across goroutines. The data moves; only one goroutine owns it at a time.
- Use **sync/atomic** (next chapter) for single words — flags and counters — where even a mutex is overkill.

If a mutex makes the code simpler, use the mutex. Idiomatic Go is not "channels everywhere"; it is "the simplest correct tool."

12. sync/atomic & Lock-Free Programming

For single machine words — a flag, a counter, a pointer swap — a full mutex is heavier than necessary. `sync/atomic` provides hardware-level atomic operations that are indivisible: no other goroutine can observe a half-completed update.

12.1 The typed atomic API (Go 1.19+)

Modern Go provides wrapper types that are safer and harder to misuse than the raw functions:

`atomic.Int32`, `atomic.Int64`, `atomic.Uint32`, `atomic.Uint64`, `atomic.Bool`, `atomic.Pointer[T]`, and `atomic.Value`.

```
type Metrics struct {
    requests atomic.Int64           // zero value ready to use
    healthy  atomic.Bool
}
func (m *Metrics) Hit()           { m.requests.Add(1) }           // atomic increment
func (m *Metrics) Count() int64 { return m.requests.Load() }      // atomic read
func (m *Metrics) SetHealthy(b bool) { m.healthy.Store(b) }      // atomic write
```

These types must not be copied after use (they contain a `noCopy` marker `go vet` understands), so hold them behind a pointer like a mutex.

12.2 The fundamental operations

- **Load / Store** — atomically read or write a word. Prevents *torn reads/writes* and establishes ordering.
- **Add** — atomic read-modify-write addition (use a negative delta to subtract).
- **Swap** — atomically set a new value and return the old one.
- **CompareAndSwap (CAS)** — the cornerstone of lock-free algorithms: "if the value is still `old`, set it to `new`; report whether you succeeded."

```
var counter atomic.Int64

// Lock-free increment-with-condition using a CAS retry loop:
func incIfBelow(limit int64) bool {
    for {
        cur := counter.Load()
        if cur >= limit {
            return false
        }
        if counter.CompareAndSwap(cur, cur+1) { // succeeds only if unchanged since Load
            return true
        }
        // someone else changed it; loop and retry
    }
}
```

The CAS retry loop is the heart of lock-free programming: read the current value, compute a new one, and atomically swap *only if nothing changed in between*. If another goroutine raced you, CAS fails and you retry. No locks, no blocking — but watch for the **ABA problem** (a value changes A→B→A and CAS wrongly succeeds) in pointer-based structures.

12.3 atomic.Pointer and atomic.Value — swapping whole snapshots

`atomic.Pointer[T]` (typed, generic) and `atomic.Value` (untyped) let you atomically publish an entire immutable snapshot — a powerful pattern for hot-read, rare-write config:

```
type Config struct{ Timeout time.Duration; Retries int }

var cfg atomic.Pointer[Config]

func init()          { cfg.Store(&Config{Timeout: time.Second, Retries: 3}) }
func Current() *Config { return cfg.Load() }                               // lock-free read
func Reload(c *Config) { cfg.Store(c) }                                   // atomic swap of the whole config
```

Readers get a consistent, never-torn `*Config` with zero locking; a writer swaps in a brand-new immutable struct. This **copy-on-write** approach beats a mutex when reads are vastly more frequent than writes. The rule: treat the pointed-to value as **immutable** — never mutate `*Current()`; build a new `Config` and `Store` it.

12.4 When atomics are the wrong tool

Atomics operate on **one word at a time**. The instant you need two related fields to change together — or a check-then-act across multiple variables that isn't expressible as a single CAS — atomics cannot give you consistency, and you need a mutex. Atomics are also notoriously easy to get subtly wrong around memory ordering. The guidance from the Go team is blunt: *use atomics for simple counters and flags; reach for channels or mutexes for anything else*. Cleverness here is rarely worth the maintenance cost.

13. The context Package

`context` is how Go programs propagate **cancellation, deadlines, and request-scoped values** across API boundaries and goroutine trees. In any networked service, it is non-negotiable: it is the standard mechanism for "stop working, the caller gave up."

13.1 Why context exists

When an HTTP request arrives, it may spawn a tree of goroutines — database calls, RPCs, fan-outs. If the client disconnects or a timeout fires, *all* of that work should stop promptly to free resources.

`context` carries a cancellation signal down the entire tree so one cancel unwinds everything.

13.2 The Context interface

```
type Context interface {
    Done() <-chan struct{}           // closed when cancelled/expired
    Err() error                     // why it's done: Canceled or DeadlineExceeded
    Deadline() (deadline time.Time, ok bool) // when it will auto-cancel, if ever
    Value(key any) any              // request-scoped value lookup
}
```

The crucial method is `Done()`: it returns a channel that is **closed** when the context is cancelled. Goroutines `select` on it exactly like a manual `done` channel — `context` is the standardized, composable version of the Chapter 10 pattern.

13.3 Deriving contexts

You never create a `Context` from nothing except at the top. You start from `context.Background()` (the root, in `main` /init/tests) or `context.TODO()` (placeholder when unsure), then **derive** child contexts:

```
ctx, cancel := context.WithCancel(parent)           // manual cancellation
ctx, cancel := context.WithTimeout(parent, 5*time.Second) // auto-cancel after duration
ctx, cancel := context.WithDeadline(parent, deadline) // auto-cancel at a time
ctx := context.WithValue(parent, key, val)           // attach a request-scoped value
```

Cancellation flows **down**: cancelling a parent cancels all descendants, but not vice versa. Each derivation forms a tree edge.

13.4 The cancel function must always be called

`WithCancel`, `WithTimeout`, and `WithDeadline` return a `cancel` function. **You must call it** — always, on every path — to release the context's resources, even if the operation finished successfully. The idiom is `defer cancel()` right after creation:


```
func fetch(ctx context.Context, url string) error {
    ctx, cancel := context.WithTimeout(ctx, 3*time.Second)
    defer cancel() // ALWAYS – prevents a context leak

    req, _ := http.NewRequestWithContext(ctx, "GET", url, nil)
    resp, err := http.DefaultClient.Do(req) // aborts if ctx expires
    if err != nil {
        return err
    }
    defer resp.Body.Close()
    return nil
}
```

`go vet` warns if a `cancel` from these functions is not called on some path — heed it; the leak is real (a timer and a goroutine linger until the parent is cancelled).

13.5 Respecting cancellation in your own code

Receiving a `ctx` obliges you to *check* it. In any loop or blocking wait, `select` on `ctx.Done()`:

```
func worker(ctx context.Context, jobs <-chan int) error {
    for {
        select {
            case <-ctx.Done():
                return ctx.Err() // Canceled or DeadlineExceeded – propagate it
            case j, ok := <-jobs:
                if !ok {
                    return nil
                }
                if err := handle(ctx, j); err != nil { // pass ctx DOWN to callees
                    return err
                }
            }
        }
    }
}
```

Functions that perform blocking I/O should accept a `ctx` and honor it (the standard library's `net`, `http`, `database/sql`, and `os` packages all do). Always pass `ctx` as the **first parameter**, named `ctx`, and never store it in a struct — thread it explicitly through calls.

13.6 context.WithValue — use sparingly

`WithValue` attaches request-scoped data (a request ID, auth principal, trace span) that crosses API boundaries. Strict discipline:

- Use it for **request-scoped** values that transit process boundaries, **not** for passing optional function parameters — that obscures your API.
- Keys must be **unexported custom types** to avoid collisions, never bare strings.

```

type ctxKey int
const requestIDKey ctxKey = 0

func WithRequestID(ctx context.Context, id string) context.Context {
    return ctx.WithValue(ctx, requestIDKey, id)
}
func RequestID(ctx context.Context) (string, bool) {
    id, ok := ctx.Value(requestIDKey).(string)
    return id, ok
}

```

`Value` does a linear walk up the context chain on every lookup, so it is not for hot-path data. Keep the payload small and immutable.

13.7 The context contract — a summary

- Pass `ctx` as the first argument; do not store it in structs.
- Only the top of the call tree creates a root (`Background`); everyone else derives.
- Always `defer cancel()`; never drop a cancel function.
- Check `ctx.Done()` in loops and blocking operations; propagate `ctx.Err()`.
- `context` is for cancellation and request scope — not a general-purpose data bag.

14. The Go Memory Model & Happens-Before

This chapter is what separates "my concurrent code seems to work" from "I can *prove* my concurrent code is correct." The Go Memory Model defines precisely when a read in one goroutine is guaranteed to observe a write from another.

14.1 The problem: without synchronization, there are no guarantees

Modern CPUs and compilers **reorder** memory operations for speed. Each goroutine may run on a different core with its own caches. Absent synchronization, there is **no guarantee** that one goroutine ever sees another's writes, or sees them in program order.

```
var a, b int
func main() {
    go func() {
        a = 1           // (1)
        b = 2           // (2)
    }()
    fmt.Println(b, a)    // may print 0 0, 0 1, 2 0, or 2 2 – ALL legal without sync
}
```

The compiler or CPU may reorder (1) and (2); the reading goroutine may observe them out of order or not at all. This is not a bug in Go — it is the reality of shared-memory hardware. The memory model tells you which orderings you can **rely on**.

14.2 Happens-before: the core relation

The memory model is defined in terms of a partial order called **happens-before**. If event *e1* happens-before *e2*, then the effects of *e1* are guaranteed visible to *e2*. The rules:

- **Within a single goroutine**, program order *is* happens-before: statements execute as written (the compiler may reorder only when unobservable to that goroutine).
- **Across goroutines**, ordering is established *only* by synchronization events. The ones you can rely on:

14.3 The synchronization guarantees

These are the concrete happens-before edges Go promises:

1. **Goroutine creation.** The `go` statement happens-before the start of the new goroutine. So the new goroutine sees everything the parent did before launching it.
2. **Goroutine completion does *not* synchronize by itself.** A goroutine exiting is *not* observable without explicit synchronization — you need a channel, `WaitGroup`, etc. (This is why you can't just

"launch and read a result variable.")

3. **Channel send happens-before the corresponding receive completes.** A send on a channel happens-before the receive that obtains that value finishes. This is the workhorse: passing a value through a channel publishes *all* prior writes of the sender to the receiver.
4. **Channel close happens-before a receive that observes the closed state** (the zero-value, `ok=false` receive).
5. **Unbuffered channel: the receive happens-before the send completes.** (The rendezvous synchronizes both directions.) For buffered channels, the k -th receive happens-before the $(k+C)$ -th send, where C is capacity.
6. **Mutex.** For a `sync.Mutex` / `RWMutex`, the n -th `Unlock` happens-before the $(n+1)$ -th `Lock` returns. Releasing a lock publishes your writes to whoever acquires it next.
7. **sync.Once.** The single execution of `f` inside `once.Do(f)` happens-before any `once.Do(f)` call returns. Every caller sees the initialized result.
8. **WaitGroup.** All `Done()` calls happen-before the `Wait()` they release.
9. **atomic operations** establish ordering: an atomic write happens-before a subsequent atomic read of the same location that observes it (sequentially consistent semantics for the typed atomics).

14.4 Why "it worked on my machine" is dangerous

A data-racy program may appear correct for years — on one CPU architecture, one compiler version, under light load — then fail in production after a deploy or under contention. x86 has a relatively *strong* memory model that hides many races; ARM (Apple Silicon, Graviton, mobile) is *weaker* and exposes them. Code that "works" on your x86 laptop can corrupt data on an ARM server. **Never** reason from observed behavior; reason from the happens-before rules. And always test under the race detector (Chapter 24).

14.5 The fixed example

To make the Chapter 14.1 example correct, introduce a happens-before edge — here, a channel:

```
var a, b int
func main() {
    done := make(chan struct{})
    go func() {
        a = 1
        b = 2
        close(done)           // writes to a,b happen-before this close...
    }()
    <-done                    // ...which happens-before this receive returns
    fmt.Println(b, a)         // GUARANTEED to print 2 1
}
```

The `close`/receive pair creates the ordering edge: every write before `close(done)` is visible after `<-done` returns. Now the output is deterministic.

14.6 The one rule to remember

If two goroutines access the same memory and at least one is a write, you **MUST** synchronize them — with a channel, a mutex, or an atomic. There are no exceptions.

Read-only sharing (no writes after publication) is safe. Everything else requires an explicit happens-before edge. Internalizing this single sentence prevents the entire category of data-race bugs.

Part III — Concurrency Patterns

These are the composable building blocks of production Go. Each is small; mastery is knowing which to reach for and how they combine. Throughout, two invariants recur: **every goroutine must have a clear exit**, and **cancellation must propagate**. Patterns that violate either leak goroutines.

15. The Pipeline Pattern

A **pipeline** is a series of stages connected by channels, where each stage is a group of goroutines running the same function. Each stage receives values from an inbound channel, transforms them, and sends results on an outbound channel. Data flows like an assembly line.

15.1 Anatomy of a pipeline

```
// Stage 1: a "generator" / source - produces values.
func gen(nums ...int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)           // source owns and closes its output
        for _, n := range nums {
            out <- n
        }
    }()
    return out
}

// Stage 2: square each number.
func sq(in <-chan int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)           // drains until upstream closes
        for n := range in {
            out <- n * n
        }
    }()
    return out
}

// Compose: gen -> sq -> sq -> print
func main() {
    for v := range sq(sq(gen(2, 3, 4))) {
        fmt.Println(v)             // 16 81 256
    }
}
```

Each stage: (1) receives on an inbound channel until it is closed, (2) sends on an outbound channel, (3) **closes the outbound channel when done**. Closing propagates downstream and tears the pipeline down cleanly from the source. The directional channel types (`<-chan int`) document and enforce the data direction.

15.2 Why pipelines are powerful

- **Composability:** stages are independent functions you can rearrange, reuse, and test in isolation.
- **Natural backpressure:** with unbuffered (or small-buffered) channels, a slow downstream stage blocks the upstream one, automatically throttling production to consumption rate. No queue grows unbounded.

- **Parallelism for free:** a CPU-bound stage can be replaced by a fan-out of N identical goroutines (Chapter 16) without touching neighboring stages.

15.3 Adding cancellation

The naive pipeline leaks goroutines if the consumer stops early: a stage blocked on `out <- n` with no receiver never returns. Real pipelines thread a `context` (or `done` channel) and `select` on it for **every** send:

```
func sq(ctx context.Context, in <-chan int) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)
        for n := range in {
            select {
            case out <- n * n:
            case <-ctx.Done():      // consumer left / deadline hit → unwind
                return
            }
        }
    }()
    return out
}
```

Now cancelling the context unblocks every stage's send and the whole pipeline drains and exits.

This `select`-on-send is the single most important habit for leak-free pipelines. Pair it with the upstream stages also watching `ctx.Done()` on their sends, and the entire chain collapses promptly on cancel.

16. Fan-Out / Fan-In

When one pipeline stage is the bottleneck, **fan-out** distributes its work across multiple goroutines reading the same channel; **fan-in** merges their results back into one channel. Together they parallelize a stage transparently.

16.1 Fan-out: multiple readers, one source

Multiple goroutines reading from the *same* channel is safe and is the natural load-balancer — each value is received by exactly one reader, and the runtime hands work to whichever goroutine is free:

```
func main() {
    in := gen(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)

    // Fan-out: start 3 workers all reading from `in`.
    c1 := sq(in)
    c2 := sq(in)
    c3 := sq(in)

    // Fan-in: merge c1, c2, c3 into one channel.
    for v := range merge(c1, c2, c3) {
        fmt.Println(v)
    }
}
```

16.2 Fan-in: merging channels with a WaitGroup

The merge function launches one goroutine per input channel, all forwarding to a shared output, and closes the output **once all inputs are drained** — coordinated by a `WaitGroup`:

```
func merge(cs ...<-chan int) <-chan int {
    out := make(chan int)
    var wg sync.WaitGroup

    forward := func(c <-chan int) {
        defer wg.Done()
        for v := range c {           // drain this input fully
            out <- v
        }
    }

    wg.Add(len(cs))
    for _, c := range cs {
        go forward(c)
    }

    // Closer goroutine: wait for all forwarders, THEN close out.
    go func() {
        wg.Wait()
        close(out)                // exactly one closer, after all sends are done
    }()
}
```

```
    return out  
}
```

This is the canonical solution to the "multiple senders, who closes?" problem from Chapter 9: nobody sends after `wg.Wait()` returns, so closing there is provably safe. The `WaitGroup` counts senders; a dedicated goroutine closes once the count hits zero.

16.3 Bounded fan-out

Unbounded fan-out (one goroutine per item over a huge stream) can exhaust memory or downstream capacity. In practice you fan out to a **fixed** number of workers — which is exactly the worker pool (next chapter). Choose the worker count from the workload: roughly `GOMAXPROCS` for CPU-bound work, and higher (tuned by measurement) for I/O-bound work where goroutines spend most of their time blocked.

17. The Worker Pool Pattern

A **worker pool** processes a stream of jobs with a *fixed* number of goroutines. It is the most common concurrency pattern in production Go — it bounds resource usage (no goroutine explosion), provides natural load balancing, and is trivial to reason about.

17.1 The canonical worker pool

```
func worker(id int, jobs <-chan int, results chan<- int, wg *sync.WaitGroup) {
    defer wg.Done()
    for j := range jobs {           // each worker pulls jobs until `jobs` is closed
        results <- j * j           // do the work, send the result
    }
}

func main() {
    const numWorkers = 4
    jobs := make(chan int, 100)
    results := make(chan int, 100)
    var wg sync.WaitGroup

    // Start a fixed pool of workers.
    for w := 1; w <= numWorkers; w++ {
        wg.Add(1)
        go worker(w, jobs, results, &wg)
    }

    // Send jobs, then close so workers' `range` terminates.
    for j := 1; j <= 20; j++ {
        jobs <- j
    }
    close(jobs)

    // Close results once all workers are done.
    go func() {
        wg.Wait()
        close(results)
    }()

    // Collect.
    for r := range results {
        fmt.Println(r)
    }
}
```

The structure is worth memorizing because it recurs everywhere:

1. A **fixed** number of workers all **range** over one shared **jobs** channel (this is the fan-out and the load balancer in one).
2. The producer **close**s **jobs** to signal "no more work," ending every worker's loop.
3. A **WaitGroup** tracks worker completion; a dedicated goroutine **close**s **results** after **Wait()** so the collector's **range** ends cleanly.

4. The main goroutine drains `results`.

17.2 Why fixed workers beat goroutine-per-job

Spawning a goroutine per job is fine for *bounded* work, but for an unbounded or very large stream it has problems a pool solves:

- **Resource bounds.** N workers cap concurrent memory, open files, and DB connections. Goroutine-per-job can open ten thousand connections and exhaust the database.
- **Backpressure.** A buffered `jobs` channel of fixed size means producers block when workers fall behind — the system self-throttles instead of accumulating an unbounded backlog.
- **Predictable performance.** A fixed pool matched to cores or downstream limits avoids the thrash of oversubscription.

17.3 A production-grade pool with context and errors

Real pools honor cancellation and surface errors. This version stops promptly on context cancellation and is structured for clarity:

```
func runPool(ctx context.Context, numWorkers int, jobs <-chan Job) <-chan Result {
    results := make(chan Result)
    var wg sync.WaitGroup

    wg.Add(numWorkers)
    for i := 0; i < numWorkers; i++ {
        go func() {
            defer wg.Done()
            for {
                select {
                    case <-ctx.Done():
                        return // cancelled: stop pulling work
                    case job, ok := <-jobs:
                        if !ok {
                            return // jobs exhausted
                        }
                        res := process(ctx, job) // pass ctx down to the unit of work
                        select {
                            case results <- res:
                                case <-ctx.Done():
                                    return // don't block forever if consumer left
                        }
                }
            }
        }()
    }

    go func() { wg.Wait(); close(results) }()
    return results
}
```

For pools where workers can fail and you want first-error-cancels-all semantics, prefer `errgroup` (Chapter 21), which folds the `WaitGroup`, error collection, and context cancellation into one tool.

18. Bounded Parallelism & Semaphores

Sometimes you do not want a long-lived pool — you want to run a known set of tasks but cap how many execute **simultaneously** (e.g., "fetch 10,000 URLs, but at most 50 at a time"). A **counting semaphore** built from a buffered channel is the idiomatic solution.

18.1 A semaphore from a buffered channel

A buffered channel of capacity N is a counting semaphore: sending acquires a slot (blocks when N are taken), receiving releases one.

```
func fetchAll(ctx context.Context, urls []string) error {
    const maxConcurrent = 50
    sem := make(chan struct{}, maxConcurrent) // N tokens
    var wg sync.WaitGroup

    for _, url := range urls {
        wg.Add(1)
        go func() {
            defer wg.Done()

            select {
            case sem <- struct{}{}: // ACQUIRE a token (blocks if 50 in flight)
            case <-ctx.Done():
                return
            }
            defer func() { <-sem }() // RELEASE the token

            fetch(ctx, url)
        }()
    }
    wg.Wait()
    return nil
}
```

At most `maxConcurrent` goroutines hold a token and run the body at once; the rest block on `sem <- struct{}{}` until a slot frees. This bounds concurrency without a persistent worker pool, and it scales to launching one goroutine per task because only N are ever *active*.

18.2 golang.org/x/sync/semaphore — the weighted version

For weighted resources (a task that costs 3 units of a 10-unit budget) the official extended library provides a richer semaphore that also respects context:

```
import "golang.org/x/sync/semaphore"

sem := semaphore.NewWeighted(10) // 10 units total
if err := sem.Acquire(ctx, 3); err != nil { // acquire 3, or fail on ctx cancel
    return err
}
```

```
}  
defer sem.Release(3)
```

Use the channel form for the common unweighted case; use `x/sync/semaphore` when tasks have differing costs or you want context-aware acquisition out of the box.

18.3 Choosing the limit

The right concurrency cap is a property of the *bottleneck resource*, not of Go: the number of DB connections in your pool, the rate limit of an upstream API, available memory per task, or core count for CPU-bound work. Pick the number that keeps the constrained resource near full utilization without overrunning it — and confirm with measurement, never a guess.

19. Rate Limiting

Rate limiting caps the *frequency* of operations (requests per second), distinct from semaphores which cap *simultaneous* operations. You need it to respect upstream API quotas, protect your own services, and avoid being throttled or banned.

19.1 A simple ticker-based limiter

`time.Ticker` delivers a value on a channel at a fixed interval. Receiving from it before each operation paces you to that rate:

```
func main() {
    // Allow one operation every 200ms (= 5 per second).
    limiter := time.NewTicker(200 * time.Millisecond)
    defer limiter.Stop()

    for req := range requests {
        <-limiter.C           // wait for the next tick before proceeding
        handle(req)
    }
}
```

Simple, but it permits no bursts — each operation waits its turn even if the system was idle. For most real systems you want to allow short bursts while bounding the long-run average. That is the **token bucket**.

19.2 The token bucket: `golang.org/x/time/rate`

The standard solution is `golang.org/x/time/rate`, a token-bucket limiter. Tokens refill at rate `r` per second up to a bucket size `b` (the burst). Each operation consumes a token; an empty bucket forces waiting.

```
import "golang.org/x/time/rate"

// 10 events/sec sustained, bursts of up to 30.
limiter := rate.NewLimiter(rate.Limit(10), 30)

func handle(ctx context.Context, req Request) error {
    if err := limiter.Wait(ctx); err != nil { // block until a token is free (or ctx done)
        return err
    }
    return doWork(req)
}
```

Three ways to consume a token, for different needs:

- `Wait(ctx)` — block until a token is available or the context is cancelled. Use when you want to *throttle* and can afford to wait.

- `Allow()` — return `true` / `false` immediately; no waiting. Use to *reject/drop* over-limit requests (e.g., return HTTP 429).
- `Reserve()` — reserve a token and learn how long until it is valid, for custom scheduling.

```
if !limiter.Allow() {  
    http.Error(w, "rate limit exceeded", http.StatusTooManyRequests)  
    return  
}
```

The token bucket elegantly captures real-world intent: *"on average no more than R/sec, but tolerate a burst of B after idle."* Tune `r` to the upstream's documented limit (with margin) and `b` to how much burstiness it tolerates.

19.3 Per-client limiting

To rate-limit *each* client independently (e.g., per API key), keep a `map[string]*rate.Limiter` guarded by a mutex, creating limiters lazily and evicting idle ones. This is a common middleware pattern; the building blocks are exactly the guarded-map (Chapter 11) and the token bucket above.

20. Cancellation, Timeouts & Deadlines

Every blocking operation in a robust system must be *bounded* — able to give up. This chapter consolidates the cancellation tools into a coherent discipline, because **goroutine leaks** (Chapter 23) are overwhelmingly caused by goroutines that can block forever with no way to stop.

20.1 The three questions for every goroutine

Before you write `go`, answer:

1. **When does it stop?** Every goroutine needs a guaranteed exit condition — a closed channel, a finished `range`, a cancelled context.
2. **Who tells it to stop?** Cancellation must reach it — via `ctx.Done()`, a `done` channel, or a closed input.
3. **What if the consumer disappears?** A goroutine blocked on a send needs an escape (`select` with `ctx.Done()`), or it leaks when nobody receives.

If you cannot answer all three, the goroutine is a latent leak.

20.2 Timeout on an operation

Wrap any operation that might hang with a deadline. The two equivalent idioms:

```
// (a) context-based – preferred; propagates into ctx-aware callees.
ctx, cancel := context.WithTimeout(ctx, 5*time.Second)
defer cancel()
result, err := doWork(ctx)      // doWork must honor ctx.Done()

// (b) select + time.After – for a one-off wait on a channel you don't control.
select {
case res := <-resultCh:
    use(res)
case <-time.After(5 * time.Second):
    return errors.New("operation timed out")
}
```

Prefer (a) whenever the callee accepts a context — the timeout then actually *cancels the work* (closing sockets, aborting queries), not merely abandons the waiting goroutine. Form (b) abandons the operation: the producer goroutine may keep running and **leak** unless it too watches for cancellation. This distinction is the difference between a timeout that frees resources and one that quietly accumulates zombie goroutines.

20.3 Propagating cancellation through a tree

A single root context cancels an entire goroutine tree. The pattern in a server:

```

func handleRequest(w http.ResponseWriter, r *http.Request) {
    ctx := r.Context()           // cancelled automatically if the client disconnects

    // Derive a tighter deadline for a downstream call.
    dbCtx, cancel := context.WithTimeout(ctx, 2*time.Second)
    defer cancel()

    user, err := db.GetUser(dbCtx, id) // aborts if client leaves OR 2s elapses
    if err != nil {
        http.Error(w, err.Error(), http.StatusGatewayTimeout)
        return
    }
    // ...
}

```

`net/http` cancels `r.Context()` when the client connection closes, so honoring it everywhere downstream means a client hangup instantly frees all the server-side work it triggered. This is the payoff of threading `ctx` consistently.

20.4 The "first to finish wins" race

To take the result of whichever of several operations completes first (e.g., query multiple replicas, use the fastest), launch them with a shared cancellable context and select the first result, then cancel the rest:

```

func fastest(ctx context.Context, replicas []string) (Result, error) {
    ctx, cancel := context.WithCancel(ctx)
    defer cancel() // cancel the losers when we return

    results := make(chan Result, len(replicas)) // buffered so losers don't block on send
    for _, r := range replicas {
        go func() {
            if res, err := query(ctx, r); err == nil {
                select {
                case results <- res:
                case <-ctx.Done():
                }
            }
        }()
    }

    select {
    case res := <-results:
        return res, nil // first success; defer cancel() stops the rest
    case <-ctx.Done():
        return Result{}, ctx.Err()
    }
}

```

Two details make this leak-free: the results channel is **buffered** to `len(replicas)` so a losing goroutine's send never blocks, and `defer cancel()` signals the losers to abandon their now-pointless work. Forgetting either reintroduces a leak.

21. errgroup & Coordinated Error Handling

A `WaitGroup` waits for goroutines but ignores errors. `golang.org/x/sync/errgroup` is the production tool for running a group of goroutines that can **fail**: it waits for all of them, returns the **first non-nil error**, and (in its context form) **cancels the rest** the moment one fails. It is the single most useful concurrency helper for everyday service code.

21.1 The basic Group

```
import "golang.org/x/sync/errgroup"

func fetchAll(urls []string) error {
    var g errgroup.Group

    for _, url := range urls {
        g.Go(func() error {           // each func returns an error
            resp, err := http.Get(url)
            if err != nil {
                return err           // first error is captured
            }
            return resp.Body.Close()
        })
    }
    return g.Wait()                  // waits for all; returns the first error (or nil)
}
```

`g.Go` launches a goroutine; `g.Wait` blocks until all of them return and yields the first error encountered. No manual `WaitGroup`, no error channel, no select — the boilerplate is gone.

21.2 WithContext: first error cancels the rest

The form you will use most. `errgroup.WithContext` returns a group and a derived context that is **cancelled as soon as any goroutine returns an error** (or all succeed). Pass that context into each task so siblings stop wasting work on a request that has already failed:

```
func processAll(ctx context.Context, ids []int) error {
    g, ctx := errgroup.WithContext(ctx) // ctx is cancelled on first error

    results := make([]Result, len(ids))
    for i, id := range ids {
        i, id := i, id // capture (pre-1.22; harmless after)
        g.Go(func() error {
            res, err := fetch(ctx, id) // honors ctx → aborts when a sibling fails
            if err != nil {
                return fmt.Errorf("id %d: %w", id, err)
            }
            results[i] = res // safe: each goroutine writes a DISTINCT index
            return nil
        })
    }
}
```

```

    if err := g.Wait(); err != nil {
        return err
    }
    return process(results)
}

```

Two things make this clean and correct: the derived `ctx` propagates the first failure to every sibling, and each goroutine writes to a **distinct slice index** — disjoint memory needs no lock (a key insight: parallel writes to *different* elements of the same slice are race-free, because they touch different memory).

21.3 SetLimit: a bounded errgroup

Modern `errgroup` includes `SetLimit(n)`, which caps how many goroutines run concurrently — folding the semaphore of Chapter 18 directly into the group. This makes `errgroup` a complete, bounded, error-aware, cancellable worker mechanism in a handful of lines:

```

g, ctx := errgroup.WithContext(ctx)
g.SetLimit(10) // at most 10 concurrent goroutines
for _, item := range items {
    g.Go(func() error { return handle(ctx, item) }) // blocks here when 10 are in flight
}
err := g.Wait()

```

`g.Go` blocks when the limit is reached, so you can iterate over a huge slice while keeping only `n` goroutines active. For most "do these tasks concurrently, bounded, and stop on first error" needs, this is the entire pattern — reach for it before hand-rolling pools.

21.4 When NOT to use errgroup

If you need *all* errors (not just the first), `errgroup` is the wrong fit — collect into a slice guarded by a mutex, or use `errors.Join` to combine them. And if failures should *not* cancel siblings (you want every task to run regardless), use the plain `errgroup.Group` (not `WithContext`) or a `WaitGroup` with explicit error aggregation.

22. Generators & Iterators

A **generator** is a goroutine that produces a stream of values on a channel, letting a consumer pull them lazily with `range`. It is the source stage of a pipeline and a clean way to model "produce values on demand."

22.1 The channel generator

```
// An infinite generator of Fibonacci numbers – lazy, bounded by the consumer.
func fib(ctx context.Context) <-chan int {
    out := make(chan int)
    go func() {
        defer close(out)
        a, b := 0, 1
        for {
            select {
                case out <- a:           // blocks until the consumer wants the next value
                    a, b = b, a+b
                case <-ctx.Done():
                    return               // consumer stopped → don't leak this goroutine
            }
        }
    }()
    return out
}

func main() {
    ctx, cancel := context.WithCancel(context.Background())
    defer cancel()

    ch := fib(ctx)
    for i := 0; i < 10; i++ {
        fmt.Println(<-ch)               // pull 10 values; generator pauses between each
    }
    // cancel() (via defer) tells the generator to stop and exit cleanly.
}
```

The generator runs only as fast as the consumer pulls — the unbuffered channel provides lazy evaluation and backpressure. An *infinite* generator is perfectly safe here **only because** it watches `ctx.Done()`; without that, ending the consumer loop would leak the goroutine forever blocked on `out <- a`. This is the same leak-prevention rule from Chapter 20, applied to sources.

22.2 Go 1.23+ range-over-function iterators

Since Go 1.23, the language supports **range-over-function** iterators, often a better fit than channel generators for in-memory sequences: no goroutine, no channel overhead, and the iteration is synchronous and leak-free by construction. You write a function matching `iter.Seq[T]` and `range` over it directly:

```

import "iter"

func Fib() iter.Seq[int] {
    return func(yield func(int) bool) {
        a, b := 0, 1
        for {
            if !yield(a) {           // yield returns false when the consumer breaks
                return              // clean stop – no goroutine to leak
            }
            a, b = b, a+b
        }
    }
}

func main() {
    for v := range Fib() {          // range directly over the function
        if v > 100 {
            break                   // break makes yield return false → generator stops
        }
        fmt.Println(v)
    }
}

```

When to use which: function iterators (`iter.Seq`) are now the idiomatic choice for *synchronous, in-memory* sequences — they are cheaper and cannot leak. Reserve **channel generators** for genuinely *concurrent* production: when values arrive from I/O, from another goroutine, or must be produced in the background while the consumer does other work. If there is no concurrency requirement, prefer the function iterator.

Part IV — Mastery: Pitfalls, Debugging & Performance

23. Common Concurrency Bugs

Knowing the patterns is half of mastery; the other half is recognizing how they fail. These are the bugs you *will* write — learn their signatures so you spot them in review and in production.

23.1 The loop variable capture (the classic)

For over a decade, the most common Go concurrency bug was capturing a loop variable in a goroutine closure. Pre-Go 1.22, the loop variable was a *single* variable reused each iteration, so all goroutines saw its final value:

```
// PRE-1.22 BUG:
for _, v := range items {
    go func() {
        fmt.Println(v)           // BUG: likely prints the LAST item N times
    }()
}
```

The pre-1.22 fix was to shadow the variable per iteration: `v := v` inside the loop, or pass it as an argument `go func(v Item){...}(v)`.

Go 1.22 changed the semantics: loop variables are now **per-iteration**, so this specific bug is fixed in modern Go — the snippet above now works correctly. But you must know it exists because (a) you will read pre-1.22 code, (b) modules built for older Go versions retain the old behavior, and (c) the same capture trap still applies to *non-loop* closures sharing a variable. When in doubt, pass values as explicit arguments.

23.2 Data races

A **data race** occurs when two goroutines access the same memory concurrently and at least one writes, without synchronization (Chapter 14). The result is undefined: torn values, lost updates, corruption, or crashes — often nondeterministic and load-dependent.

```
// RACE: many goroutines increment an unguarded int.
counter := 0
var wg sync.WaitGroup
for i := 0; i < 1000; i++ {
    wg.Add(1)
    go func() { defer wg.Done(); counter++ }() // counter++ is read-modify-write: NOT atomic
}
wg.Wait()
fmt.Println(counter) // almost never 1000 - updates are lost
```

`counter++` is three steps (load, add, store); concurrent increments interleave and clobber each other. Fixes: a `sync.Mutex`, or `atomic.Int64.Add(1)`, or routing increments through a channel to a single owner. **Always run tests with `-race` (Chapter 24)** — it catches these automatically.

23.3 Deadlocks

A **deadlock** is when goroutines wait on each other in a cycle, and none can proceed. Go's runtime detects the *total* deadlock (every goroutine blocked) and panics: `fatal error: all goroutines are asleep - deadlock!`. Partial deadlocks (some goroutines stuck, others running) are *not* detected and are more insidious.

```
// DEADLOCK: unbuffered send with no receiver.
ch := make(chan int)
ch <- 1           // blocks forever; no goroutine will ever receive
fmt.Println(<-ch) // unreachable
```

Classic causes: sending on an unbuffered channel with no concurrent receiver; forgetting to `close` a channel a `range` is waiting on; lock-ordering cycles (goroutine A holds lock 1 wants 2, B holds 2 wants 1). **Prevent lock-ordering deadlocks by always acquiring multiple locks in a single, globally consistent order.**

23.4 Goroutine leaks

A **goroutine leak** is a goroutine that never terminates — blocked forever on a channel or lock, holding memory and resources. Leaks are silent: the program keeps running while goroutine count and memory climb until OOM.

```
// LEAK: goroutine blocks forever on send because the consumer stopped early.
func leak() {
    ch := make(chan int)
    go func() {
        ch <- 42           // blocks forever if nobody receives
    }()
    return                 // function returns; goroutine is stranded
}
```

The fixes are the entire content of Chapter 20: give every goroutine a guaranteed exit, `select` on `ctx.Done()` for sends, and buffer channels where producers must not block. **Detect leaks** by watching `runtime.NumGoroutine()` over time, dumping the goroutine profile (Chapter 25), or using `go.uber.org/goleak` in tests to fail the suite when a test leaves goroutines behind.

23.5 Closing a channel from the wrong place

Sending on or closing a closed channel **panics** (Chapter 9). Two failure modes dominate: a receiver closing the channel (it cannot know if the sender is done), and multiple senders each trying to close. The discipline — *the sole owner closes, closing flows downstream, multiple senders coordinate via WaitGroup before a single closer fires* — prevents both. When unsure who should close, ask: "who is the single writer?" That goroutine, and only that one, closes.

23.6 Forgetting that a panic kills every goroutine

Restating Chapter 7.3 because it causes real outages: an unrecovered panic in *any* goroutine crashes the **whole process**, and you cannot recover it from elsewhere. Long-lived workers that run untrusted or fallible code should guard themselves:

```
func safeWorker(jobs <-chan Job) {
    defer func() {
        if r := recover(); r != nil {
            log.Printf("worker recovered from panic: %v", r)
            // optionally restart, or surface via an error channel
        }
    }()
    for j := range jobs {
        process(j)      // a panic here is contained to this goroutine
    }
}
```

Do **not** sprinkle `recover` everywhere — it hides bugs. Use it at goroutine boundaries and server request handlers (where `net/http` already recovers per-request), and let genuinely impossible states crash loudly in development.

23.7 Mutex copied by value

Copying a `sync.Mutex` (or any struct containing one) after use yields two *independent* locks that protect nothing — a silent race. This happens when you pass a struct by value or store it in a map/slice.

```
type Counter struct{ mu sync.Mutex; n int }
func (c Counter) Inc() { c.mu.Lock(); c.n++; c.mu.Unlock() } // BUG: value receiver copies the mutex
```

The fix is always a **pointer receiver** and passing such structs by pointer. `go vet` reliably catches mutex copies — run it in CI.

24. The Race Detector

Go ships a **race detector** — arguably the most valuable concurrency tool in any mainstream language. It instruments memory accesses at runtime and reports when two goroutines touch the same location concurrently without synchronization, with at least one write. If you take one practical habit from this guide, make it this: **run your tests with `-race`**.

24.1 Using it

Add the `-race` flag to any build, run, or test command:

```
go test -race ./...      # run the whole test suite under the detector (do this in CI)
go run -race main.go     # run a program with detection
go build -race -o app .  # build an instrumented binary for staging
```

When a race occurs, the detector prints both stacks — the conflicting read and write, the goroutines involved, and where each was created:

```
=====
WARNING: DATA RACE
Write at 0x00c0000b4010 by goroutine 7:
    main.main.func1()
        /app/main.go:11 +0x44
Previous read at 0x00c0000b4010 by main goroutine:
    main.main()
        /app/main.go:14 +0x90
Goroutine 7 (running) created at:
    main.main()
        /app/main.go:10 +0x7c
=====
```

This tells you exactly which two lines raced and which goroutine created the offender — usually enough to find and fix the bug immediately.

24.2 What it can and cannot do

Understand the guarantees precisely:

- **No false positives.** If the detector reports a race, it is a *real* race — always. Never dismiss a report.
- **It only finds races that actually execute.** The detector is dynamic: it observes the code paths your test exercises on this run. A race on an untested path goes unseen. This is why *test coverage of concurrent paths matters* — write tests that actually run goroutines under load.
- **Cost.** Instrumentation slows execution roughly 2-20× and increases memory ~5-10×. Run `-race` in tests, CI, and staging — not in latency-sensitive production. Some teams run a small `-race` canary in production to catch rare paths.

24.3 Making races reproducible

Races are nondeterministic, so a clean `-race` run does not *prove* correctness — it only means none triggered this time. Improve your odds of catching them:

- Run tests with `-race -count=N` to repeat, and `-cpu=1,2,4` to vary parallelism.
- Increase `GOMAXPROCS` and add load/stress tests that spawn many goroutines.
- Use `t.Parallel()` in tests so the suite itself exercises concurrency.

```
go test -race -count=10 -cpu=4 ./...
```

The detector plus stress is the closest thing Go gives you to a proof; lean on it hard. Treat any `-race` failure in CI as a build-breaking bug, never a flake to retry.

25. Profiling & Tracing Concurrent Programs

When concurrent code is slow, leaks, or deadlocks, guessing wastes time. Go's built-in profiling (`pprof`) and execution tracer give you ground truth.

25.1 pprof: CPU, memory, and goroutine profiles

The `runtime/pprof` package and the `net/http/pprof` HTTP handlers expose live profiles. For a long-running service, import the HTTP handlers and visit the endpoints:

```
import _ "net/http/pprof"           // registers /debug/pprof/* on the default mux

func main() {
    go func() {
        log.Println(http.ListenAndServe("localhost:6060", nil)) // pprof server
    }()
    // ... your application ...
}
```

Then collect and analyze profiles with `go tool pprof`:

```
# 30-second CPU profile – where is time spent?
go tool pprof http://localhost:6060/debug/pprof/profile?seconds=30

# Heap profile – what is allocating / leaking memory?
go tool pprof http://localhost:6060/debug/pprof/heap

# Goroutine profile – how many goroutines, and where are they stuck?
go tool pprof http://localhost:6060/debug/pprof/goroutine

# Mutex and blocking profiles – where is contention?
go tool pprof http://localhost:6060/debug/pprof/mutex
go tool pprof http://localhost:6060/debug/pprof/block
```

Inside `pprof`, use `top` (hottest functions), `list <func>` (line-level detail), and `web` (a visual call graph; needs Graphviz). For concurrency specifically, the **goroutine profile is your leak detector**: a steadily climbing count, with thousands of goroutines parked at the same channel-receive line, points straight at the leak's source.

25.2 The execution tracer

For questions `pprof` cannot answer — *why is parallelism poor? where are goroutines blocking? is the scheduler the bottleneck?* — use the execution tracer, which records goroutine scheduling, blocking, GC, and syscall events on a timeline:

```
import "runtime/trace"

f, _ := os.Create("trace.out")
defer f.Close()
trace.Start(f)
defer trace.Stop()
// ... run the workload you want to analyze ...
```

Then open the interactive viewer:

```
go tool trace trace.out      # opens a browser UI: goroutine timelines, scheduler, GC
```

The tracer reveals what nothing else does: goroutines serializing on a lock when you expected parallelism, long GC pauses, a P sitting idle while work waits in a queue, or syscalls blocking Ms. It is the definitive tool for diagnosing "I added goroutines but it didn't get faster."

25.3 Benchmarks with -race and -bench

Microbenchmark concurrent code with the testing package's `B` type, and use `b.RunParallel` to measure under contention:

```
func BenchmarkCounter(b *testing.B) {
    var c SafeCounter
    b.RunParallel(func(pb *testing.PB) {
        for pb.Next() {
            c.Inc()           // measures throughput under GOMAXPROCS-way contention
        }
    })
}
```

```
go test -bench=. -benchmem      # include allocation stats
go test -bench=. -cpu=1,4,8     # see how throughput scales with parallelism
```

If a benchmark fails to speed up (or *slows down*) as `-cpu` rises, you have found contention — a lock too coarse, or false sharing. Confirm with the mutex/block profile and the tracer.

26. Performance Tuning for Concurrent Code

Concurrency is not automatically fast. Adding goroutines can make code *slower* through contention, scheduling overhead, and cache effects. Tuning is empirical: measure (Chapter 25), change one thing, measure again.

26.1 More goroutines is not more speed

Each goroutine is cheap but not free: it has a stack, scheduling cost, and competes for `GOMAXPROCS` Ps. For CPU-bound work, exceeding the core count adds context-switching with no parallelism gain. The classic mistakes:

- **Oversubscription** — spawning thousands of CPU-bound goroutines for an N-core machine. Cap at roughly `GOMAXPROCS` for CPU-bound work (worker pool, Chapter 17).
- **Goroutine-per-tiny-task** — when the task is shorter than the scheduling overhead, the coordination costs more than the work. Batch small tasks.
- **Unbounded fan-out** — let it rip over a huge stream and you exhaust memory or downstream limits. Always bound (Chapter 18).

For **I/O-bound** work the calculus flips: goroutines spend most of their time blocked, so far more than `GOMAXPROCS` is correct — the right number is set by the downstream resource (connection pool size, upstream rate limit), found by measurement.

26.2 Lock contention

A mutex serializes its critical section; under heavy contention, goroutines spend their time *waiting* for the lock instead of working, and adding cores stops helping. Strategies, in order of preference:

- **Shrink the critical section.** Do expensive work (I/O, allocation, computation) *outside* the lock; hold it only for the minimal state mutation.
- **Shard the lock.** Replace one mutex over a big map with N mutexes over N map shards keyed by hash — contention drops by $\sim N$. (This is what high-performance concurrent maps do internally.)
- **Use atomics** for single-word counters/flags instead of a mutex (Chapter 12).
- **Use `RWMutex`** only for genuinely read-heavy, non-trivial critical sections — and measure, since it can be slower than a plain `Mutex` for short sections.
- **Avoid sharing entirely.** Give each goroutine its own state and combine results at the end (the `errgroup` + distinct-slice-index pattern of Chapter 21).

Find contention with the **mutex and block profiles** and the **execution tracer** — never guess which lock is hot.

26.3 False sharing

CPUs move memory in **cache lines** (typically 64 bytes). If two goroutines on different cores update *different* variables that happen to share a cache line, each write invalidates the other core's cached line — "false sharing" — silently destroying performance even though there is no logical contention. The fix is **padding** to push hot per-goroutine fields onto separate cache lines:

```
type paddedCounter struct {
    n    atomic.Int64
    _    [56]byte           // pad so each counter occupies its own 64-byte cache line
}
counters := make([]paddedCounter, runtime.NumCPU()) // per-CPU counters, no false sharing
```

This is an advanced optimization — apply it only when profiling shows per-core counters or array-of-structs hot paths underperforming.

26.4 Reducing allocations and GC pressure

Garbage collection competes with your goroutines for CPU and can introduce latency. In hot concurrent paths:

- **Reuse buffers** with `sync.Pool` (Chapter 11) instead of allocating per operation.
- **Preallocate slices** with `make([]T, 0, n)` when you know the size, to avoid repeated `append` reallocations.
- **Avoid unnecessary pointer/interface boxing**, which forces heap escapes; check with `-gcflags=-m`.
- **Tune the GC** via `GOGC` (higher = less frequent GC, more memory) or `GOMEMLIMIT` (a soft memory ceiling, Go 1.19+) when you understand your memory budget.

Always confirm a GC problem with the heap profile and the tracer's GC view *before* optimizing — premature allocation-tuning obscures code for no measured benefit.

26.5 The tuning discipline

1. **Establish a benchmark** that reflects real load (`go test -bench`, `b.RunParallel`).
2. **Profile** to find the actual bottleneck (`pprof`: CPU, heap, mutex, block; `trace` for scheduling).
3. **Change one thing**. Shrink a lock, bound a fan-out, pool a buffer.
4. **Re-measure**. Keep the change only if the numbers improved.
5. **Repeat** until you hit your target, then stop. Untargeted "optimization" usually adds complexity and bugs while making nothing faster.

The cardinal rule: *measure, don't guess*. Intuition about concurrent performance is unreliable even for experts — the profiler is the source of truth.

27. Best Practices & Idioms Reference

A dense, scannable distillation of the whole guide. When you are writing concurrent Go, run down these lists.

27.1 The Go concurrency proverbs

- *Don't communicate by sharing memory; share memory by communicating* — but use a mutex when it is simpler.
- *Concurrency is not parallelism.* Structure with concurrency; the runtime supplies parallelism.
- *Channels orchestrate; mutexes serialize.* Pick the tool that fits the job.
- *The bigger the interface, the weaker the abstraction.* Keep interfaces tiny.
- *Don't just check errors, handle them gracefully.*
- *A little copying is better than a little dependency.*
- *Clear is better than clever.* Concurrency magnifies the cost of cleverness.

27.2 Goroutine lifecycle rules

- Before writing `go`, answer: **when does it stop, who stops it, what if the consumer leaves?**
- Never start a goroutine without knowing how it ends. Every goroutine needs a guaranteed exit.
- Pass `context.Context` to anything that blocks; `select` on `ctx.Done()` in every loop and every channel send that could block.
- Make the *caller* responsible for a goroutine's lifetime where possible — return a channel or accept a context rather than launching detached background work.
- Wrap long-lived/fallible worker goroutines in a deferred `recover` — an unrecovered panic anywhere crashes the whole process.

27.3 Channel rules

- Default to **unbuffered**; add a buffer only when you can justify its exact size.
- **Only the sender closes**, and only with a single owner. Receivers never close. Multiple senders coordinate via `WaitGroup` before one dedicated closer fires.
- Closing is a **broadcast** — use `close(done)` to cancel N receivers at once.
- Sending on / closing a closed channel **panics**; closing/operating on a nil channel blocks forever (use nil to disable a `select` case).
- `for range ch` ends only when `ch` is closed *and* drained — an unclosed channel blocks the range forever.
- Use `chan struct{}` for pure signaling; use directional types (`chan<-`, `<-chan`) in signatures to document and enforce flow.

27.4 Synchronization rules

- `defer mu.Unlock()` immediately after `mu.Lock()`. Keep critical sections tiny; never do I/O or block under a lock.
- Go mutexes are **not reentrant**. Never copy a `sync` type — always use pointer receivers; run `go vet`.
- `WaitGroup`: Add **before** `go`, `Done` deferred, don't reuse before `Wait` returns.
- Use `sync/atomic` for single-word counters and flags; treat values published via `atomic.Pointer` / `Value` as immutable (copy-on-write).
- `sync.Map` only for write-once-read-many or disjoint-key access; otherwise a `Mutex`-guarded plain map is faster and clearer.
- Choose: **mutex** to guard shared memory, **channel** to transfer ownership/signal, **atomic** for a single word.

27.5 Cancellation & error rules

- Pass `ctx` as the **first** argument; never store it in a struct.
- Always `defer cancel()` for `WithCancel` / `WithTimeout` / `WithDeadline` — heed the `go vet` lost-cancel warning.
- Bound every blocking operation with a timeout/deadline; prefer context-based timeouts that actually abort the work over `select` + `time.After` that merely abandons it.
- Use `errgroup.WithContext` for "run these concurrently, stop on first error"; add `SetLimit(n)` to bound it. Use `errors.Join` when you need *all* errors.
- Wrap errors with `%w` and context as you propagate; match with `errors.Is` / `errors.As`, never `==` or string comparison.

27.6 The pre-commit checklist for concurrent code

1. Does every goroutine have a guaranteed exit? (No leaks.)
2. Is every shared mutable access synchronized? (No races.)
3. Does cancellation propagate end-to-end via context?
4. Is every blocking operation bounded by a timeout/deadline?
5. Is channel-closing owned by a single sender, flowing downstream?
6. Are critical sections minimal and lock-free of I/O?
7. Does `go vet ./...` pass? Does `go test -race ./...` pass (ideally `-count=N -cpu=1,2,4`)?
8. Have you profiled before optimizing, and bounded any fan-out?

If all eight are satisfied, your concurrent Go is in excellent shape.

27.7 The essential toolchain commands

```
go vet ./...           # static checks: copied locks, lost cancels, bad printf, ...
go test -race ./...     # the race detector – run in CI, non-negotiable
```

```
go test -race -count=10 -cpu=1,2,4 ./... # stress concurrency to surface flaky races
go build -gcflags='-m' ./... # escape analysis (heap vs stack)
go tool pprof <profile-url> # CPU/heap/goroutine/mutex/block analysis
go tool trace trace.out # scheduler/blocking/GC timeline
go test -bench=. -benchmem -cpu=1,4,8 # benchmark scaling and allocations
golangci-lint run # aggregate linters (errcheck, staticcheck, etc.)
```

Closing: The Path to Mastery

You now have the complete map: the language semantics that make concurrency reasoned-about rather than guessed-at (Part I), the primitives and the formal memory model that govern them (Part II), the patterns production systems are built from (Part III), and the tools and discipline to debug and tune them (Part IV).

Mastery, though, is not absorbed from a document — it is *built* through deliberate practice. A concrete progression:

1. **Re-implement every pattern from memory.** Pipeline, fan-out/fan-in, worker pool, semaphore, rate limiter, `errgroup` task runner. Type them without looking. When you can derive them rather than recall them, they are yours.
2. **Break things on purpose.** Write the racy counter and watch `-race` catch it. Cause a deadlock and read the runtime's report. Leak a goroutine and find it in the goroutine profile. Crash a process with a goroutine panic. Fixing bugs you *created* teaches faster than avoiding hypothetical ones.
3. **Read the standard library.** `net/http`, `database/sql`, `context`, and `sync` are master-class concurrent code. Study how the `http.Server` shuts down gracefully, how `context` propagates cancellation, how `sync.Once` is implemented. Imitate them.
4. **Build something real and concurrent.** A web crawler (pipelines + bounded fan-out + dedup), a job queue (worker pool + persistence), a rate-limited API client, a load tester. Real systems surface the edge cases no exercise does.
5. **Make `-race` and `go vet` reflexive.** Wire them into CI on day one. Treat every race report as a build-breaking bug. This single habit prevents the majority of production concurrency failures.
6. **Profile before you optimize, always.** Train yourself to reach for `pprof` and `go tool trace` instead of intuition. The profiler will repeatedly surprise you — that is exactly why it is indispensable.

Go's concurrency is small enough to hold entirely in your head — a handful of keywords, a handful of `sync` types, one memory model, and a set of composable patterns — yet deep enough to build planet-scale systems. That combination of *simplicity* and *power* is the whole point of the language. Hold the proverbs, respect the happens-before rules, give every goroutine an exit, and measure instead of guessing.

Concurrency is the structure; the runtime provides the parallelism; discipline provides the correctness.

Now go write some goroutines — and run them under `-race`.